Solving Constraints in a Theory of Subsequences

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1 Rules

$$
\begin{array}{lcl}
e_1 \cdot e_2 \leq x & \Longrightarrow & \exists\ x_a\ x_b.\ x = x_a \cdot x_b\ \wedge \\
 & & e_1 \leq x_a\ \wedge\ e_2 \leq x_b \\
 & & \\
e_1 \cdot e_2 \leq C & \Longrightarrow & e_1 \leq C\ \wedge\ e_2 \leq \epsilon\ \vee \\
 & & e_1 \leq \epsilon\ \wedge\ e_2 \leq C \\
 & & \\
e_1 \cdot e_2 \leq e_1' \cdot e_2' & \Longrightarrow & (\exists\ u\ v.\ e_1' = u \cdot v\ \wedge \\
 & & e_1 \leq u\ \wedge\ e_2 \leq v \cdot e_2')\ \vee \\
 & & (\exists\ u\ v.\ e_2' = u \cdot v\ \wedge \\
 & & e_1 \leq e_1' \cdot u\ \wedge\ e_2 \leq v) \\
 & & \\
x \leq e_1 \cdot e_2 & \Longrightarrow & \exists\ x_a\ x_b.\ x = x_a \cdot x_b\ \wedge \\
 & & x_a \leq e_1\ \wedge\ x_b \leq e_2 \\
 & & \\
C \leq e_1 \cdot e_2 & \Longrightarrow & C \leq e_1\ \vee\ C \leq e_2 \\
 & & \\
x \leq C & \Longrightarrow & x = \epsilon\ \vee\ x = C
\end{array}
$$